

Smartphone-based Human Behavior Task Modeling for Explainable Mental Health Detection Model

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ABSTRACT

Smartphone data-driven mental health detection approach is widely used. However, existing methods focus primarily on enhancing model performance through numerous superficial smartphone usage features that do not effectively capture human behaviors, rather than interpreting the model's explanatory power. This study proposes a methodology of smartphone-based human behavior task modeling to advance "Explainable AI (X-AI)" for detecting mental health states. Our approach utilizes in-app UI-level trajectory data to extract more fine-grained human behavior task mining and modeling. We introduced novel concepts to investigate the nature and scope of smartphone-based human behavior tasks. This study discusses how these concepts can enhance the trustworthiness and transparency of models for both users and system providers. Furthermore, we expect that our approach could reduce data collection, thereby reducing memory usage and mitigating privacy concerns by enabling on-device learning.

CCS CONCEPTS

• Human-centered computing → Mobile computing.

KEYWORDS

Human Behavior Modeling, Explainable AI, Digital Phenotyping, Mental Health Model, Data Mining, Markov Chain Model

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1 INTRODUCTION

Mental health issues among young adults disrupt daily lives, well-being, and economic prospects, contributing to lost productivity, healthcare costs, and social burdens such as suicide and welfare demands [4, 5]. Early diagnosis, monitoring, and management of disorders like depression, anxiety, substance use (e.g., alcohol use disorder), stress, emotional fluctuations, and cognitive impairment

(e.g., dementia) are crucial. Traditional methods for diagnosing mental health disorders, like surveys, brain imaging, and neuropsychological assessments, have limitations such as high costs, time consumption, and reliance on clinician observation and measurement, which can delay access to timely diagnosis and treatment [3]. In response, "Digital Phenotyping" has emerged as a revolutionary approach utilizing data from smartphones and digital devices to measure behavioral, cognitive, and psychological states indicative of mental health status [10].

Previous research in digital phenotyping for mental health diagnostics aims to optimize AI models by integrating numerous features extracted from the statistical processing of multi-modal sensor data from smartphones or wearables. These features aim to capture human behavior (e.g., sociability, physical activity), habits (e.g., mobility, sleep), and contextual factors (e.g., environment, vocal cues, location) in everyday settings [8]. They encompass a wide range of data such as location, accelerometer readings for physical activity, physiological signals, device status, and social interactions including app usage and direct interactions (e.g., clicks, typing, scrolls, swipes). Machine learning (ML) and deep learning (DL) models leverage these features to detect cognitive disorders and psychological states [8].

While using a large number of multi-modal sensor-based features with advanced ML or DL models can enhance mental health detection accuracy, it also raises concerns from an "Explainable AI (XAI)" perspective. A plethora of features can obscure model transparency, potentially turning it into a "Black Box," which undermines user trust. Therefore, recent digital phenotyping studies emphasize identifying and focusing on key features among the extensive feature set to enhance model interpretability and user trust in AI systems [2].

Smartphone app usage-related features (e.g., total time spent on smartphones or specific apps, usage frequency) are crucial for explaining models in many studies [8] as shown in levels 1 and 2 of figure 2. However, these features primarily capture surface-level aspects of human behavior and may not fully reflect complex tasks performed through smartphone apps. Current research often aggregates usage metrics at the app category level (e.g., Finance, Health, Social) to gain deeper insights into behavior patterns [7, 13] as depicted in level 3 of figure 2. Yet, because apps like Facebook or Google can serve multiple functions across different categories, accurately interpreting specific user behaviors remains challenging. Collecting features at the individual app level further complicates matters by increasing the diversity and volume of data collected, potentially raising privacy concerns and escalating the time and cost of data acquisition.



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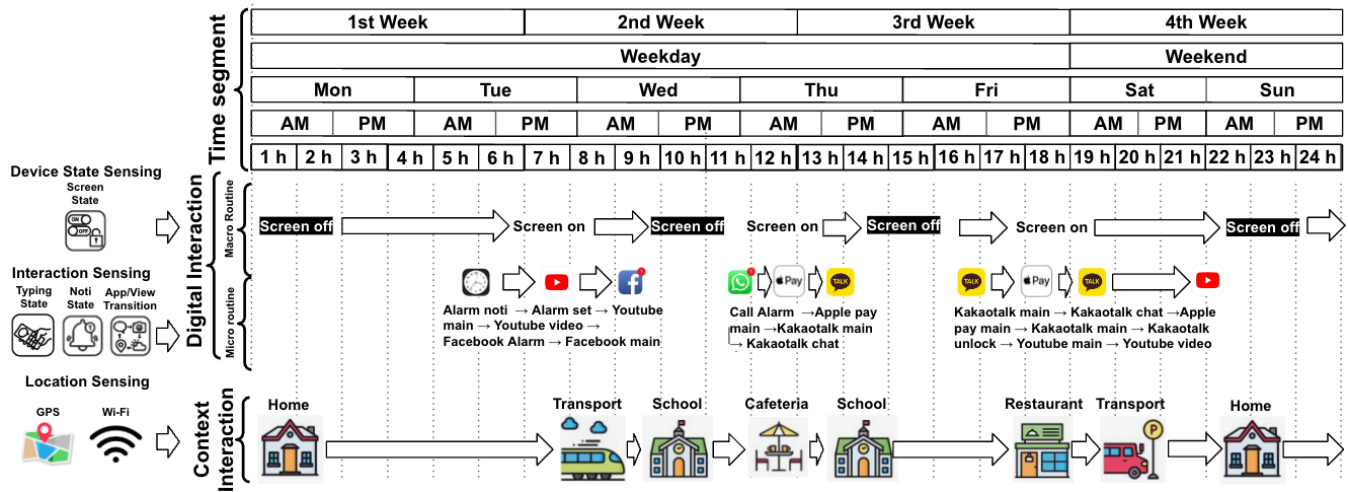


Figure 1: Smartphone Usage-based Human Behavior

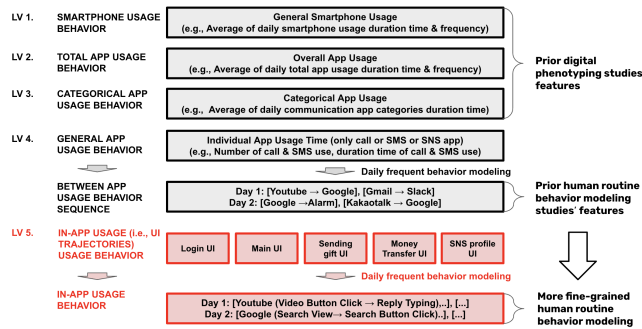


Figure 2: Level of Smartphone Usage Behavior

Identifying smartphone usage-based human behavior features at the in-app UI level as features for digital phenotyping could improve model interpretability. This approach would provide insights into specific tasks users perform across different contexts (e.g., time, location), enhancing understanding of their behaviors as shown in Figure 1. Research indicates that individuals typically engage with 9 to 10 apps daily on average [1], which reduces memory load, supports on-device learning, addresses privacy concerns, and streamlines data collection processes, thereby saving time and costs.

In this study, we explore and propose a concept for modeling human behavior using smartphone daily usage to enhance “X-AI”. To derive human behavior, our suggestion is to employ previous data mining techniques (e.g., rule and pattern mining) and modeling (Hidden Markov chain model) using smartphone usage (e.g., in-app UI trajectories) and context (e.g., location). Based on this concept, we discuss how these features can enhance the trustworthiness and transparency of the mental health model for both users and system providers.

2 HUMAN BEHAVIOR MODELING METHODOLOGY

Prior studies have conducted human routine behavior modeling based on phone usage, two context data (time and location) through smartphone passive sensing by data mining techniques such as Association Rule Mining (ARM) and Sequence Pattern Mining (SPM) [6, 9]. Other previous studies have presented human routine behavior modeling-based digital phenotyping using smartphone sensor data (e.g., location, physical activity, and app usage) to detect stress and depression, utilizing various modeling techniques such as ARM and Hidden Markov Model (HMM) methods [11, 12].

Previous studies have demonstrated the feasibility of identifying human routine behavior sequences or rules using smartphone sensor data with SPM and ARM techniques. However, as shown in Figure 1, existing approaches have primarily focused on identifying sequences or association rules between apps, neglecting multi-level patterns within and across apps. This limitation hinders the precise identification of in-app usage behavior rules or patterns critical for X-AI based on smartphone-based human behavior features. Furthermore, existing methods struggle with identifying individual smartphone-based human behavior features due to low-frequency smartphone usage behavior derived from minimum support values. To address these challenges, we propose to adopt fine-grained in-app usage data to derive human behavior. These features could be intended to enhance explainable AI.

3 DISCUSSION AND CONCLUSION

This study introduces a concept to extract smartphone usage-based human behavior tasks as explainable features for detecting mental health indicators. The smartphone usage-based human behavior features for the mental health model proposed in this study provide a more intuitive interpretation for both users and system providers compared to the features outlined in Figure 1. Traditional metrics like overall smartphone usage time/frequency or app category usage lack specificity in capturing individual human behaviors crucial

for the interpretability of the mental health model. In contrast, features based on human behaviors (e.g., task completion time, number of tasks, typing-related activities) intuitively describe everyday activities, enhancing explainable AI. For example, detailed interaction metrics such as typing speed (characters per second) and editing actions (e.g., backspaces) can differentiate between tasks like messaging and searching, offering deeper interpretability compared to broad app-based metrics.

Using smartphone-based daily routine tasks in mental health detection models offers advantages over traditional long-term and large dataset collection. More fine-grained smartphone usage behaviors are performed consistently day-to-day, making them suitable for the daily assessment of conditions like emotions, stress, depression, and dementia. Our approaches could also aid in the early detection of mild cognitive impairment. Focusing on specific tasks reduces user burden and memory usage by not collecting data from all apps, enhancing on-device learning. For providers, task-based features streamline model development by reducing data requirements. Further research is needed to validate if these features effectively distinguish stress, depression, and cognitive impairment compared to traditional methods. The proposed smartphone-based human behavior features provide a personalized explanation of smartphone use, potentially enhancing ML-based mental health detection models in personalized settings.

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